Inventory Management

MGT 207: Operations Management

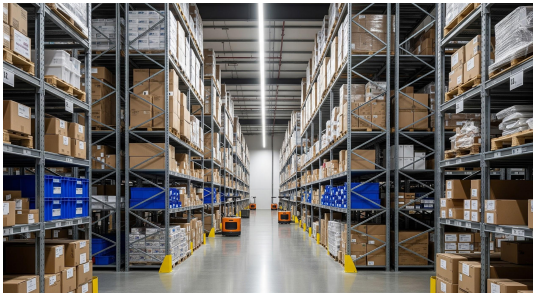
MGT 207

Inventory Management

Cycle Stock, Pipeline Stock, EOQ, Newsvendor, (Q, R) & Risk Pooling

What is Inventory & Why Executives Care

Inventory: A stock of materials or finished goods held to satisfy customer demand, decouple operations, or buffer against supply chain variability.



Strategic Role on the Balance Sheet

- **Customer Responsiveness:** Enables immediate order fulfillment without manufacturing delay.
- **Operational Decoupling:** Allows stages of production and distribution to run smoothly and independently.
- **Working Capital Commitment:** Often represents 20%–40% of a retailer's or manufacturer's invested capital.

Pressures for Small Inventories (The CFO's View)

Annual Holding (Carrying) Cost (H)

Every unit sitting on a shelf silently drains cash and Return on Invested Capital (ROIC).

Components of Holding Cost (H)

- **Cost of Capital (WACC):** Opportunity cost of cash tied up in physical stock.
- **Storage & Handling:** Warehouse space, climate control, labor, and equipment.
- **Taxes & Insurance:** Assessments on asset value and risk coverage.
- **Shrinkage & Obsolescence:** Markdowns, spoilage, damage, and technological aging.

Pressures for Large Inventories (The COO & Sales View)

Operational & Commercial Pressures

Why operations and sales executives push for deeper inventory buffers:

Key Drivers for Larger Stock

- **Customer Service & Availability:** Prevents stockouts, backorders, and lost customer goodwill.
- **Fixed Ordering & Setup Costs (S):** Ordering in large lots spreads fixed setup/shipping costs over more units.
- **Production Utilization:** Fewer changeovers stabilize factory schedules.
- **Transportation & Quantity Discounts:** Full-truckload (FTL) rates and supplier price breaks.

The Three Operational Buckets of Inventory

Why Does a Unit of Inventory Exist?

Every unit in your supply chain sits in one of three functional buckets based on the operational lever that created it:

1. Cycle Inventory

Stock resulting directly from replenishing in **batches or lot sizes (Q)** rather than one unit at a time.

Primary Driver:
Fixed ordering/setup cost (S) vs. holding cost (H).

2. Pipeline Inventory

Stock currently **in transit** between factories, distribution centers, and retail stores.

Primary Driver:
Flow/shipment rate (R) and transit lead time (L).

3. Safety Stock

Buffer stock held in excess of expected demand to absorb **uncertainty**.

Primary Driver:
Demand volatility (σ_d), lead-time volatility (σ_L), and service target.

Quantifying Cycle Stock & Pipeline Stock

1. Average Cycle Inventory (On-Hand in Warehouse: Batching Effect)

$$\textbf{Average Cycle Inventory} = \frac{Q+0}{2} = \frac{Q}{2}$$

As an order of size Q depletes uniformly to 0, average **on-hand warehouse stock** is half the lot size.

2. Pipeline Inventory (In-Transit on Trucks/Ships via Little's Law: $I = R \times T$)

$$\textbf{Pipeline Inventory} = \underbrace{R}_{\substack{\textbf{Flow / Shipment Rate} \\ \textbf{(units/wk entering transit)}}} \times \underbrace{L}_{\substack{\textbf{Transit Lead Time } (T) \\ \textbf{(wks spent on water/road)}}}$$

Note: When demand is deterministic (or 100% met in steady state), flow rate equals demand rate ($\mathbf{R} = \bar{\mathbf{d}}$).

Why Write $R \times L$ Instead of "Lead-Time Demand"?

Pipeline inventory is driven by **Little's Law ($I = R \times L$)**: how fast units flow through the shipping lane (R) times how long they sit on the boat (L). Meanwhile, the stock on hand in your warehouse while waiting L weeks is **already inside Cycle Stock ($Q/2$)**!

Mini-Case 1: The Nearshoring Dilemma

Interactive Poll & Pair Exercise (3 Minutes)

Formula Reference: Cycle Stock (Warehouse) $= \frac{Q}{2}$ | Pipeline Stock (In Transit) $= R \times L$ | Total Inventory $= \frac{Q}{2} + R \cdot L$

Executive Scenario

You are VP of Supply Chain sourcing precision actuator motors (**flow rate R = 70 units/week**, unit value $100).

Terms are **FOB Origin**: you own and pay for inventory the moment it leaves the supplier's dock.

Live Poll: Net Working Capital Impact?

Switching to the Nearshore Supplier:

- **(A)** Ties up **$3,500 more** cash in inventory
- **(B)** **Breaks even** on total inventory
- **(C)** Frees up **$3,500 in cash**

Current Offshore Supplier

- **Lot Size (Q):** 280 units
- **Flow Rate (R):** 70 units/week
- **Transit Lead Time (L):** 3 weeks

Proposed Nearshore Supplier

- **Lot Size (Q):** 350 units *(larger minimum batch)*
- **Flow Rate (R):** 70 units/week
- **Transit Lead Time (L):** 2 weeks *(faster transit)*

Safety Stock: The Buffer Against Uncertainty

What is Safety Stock?

Extra inventory held above expected demand to protect against **demand volatility (σ_d)** and **supplier lead-time variability (σ_L)**.

The Executive Trade-off

- **Too little safety stock:** Frequent stockouts, expedited freight, and lost customers.
- **Too much safety stock:** Bloated warehouses and dead working capital.

Up Next: The Master Roadmap

How do we choose the right mathematical model to manage Cycle, Pipeline, and Safety Stock?

The Master 2 × 2 Inventory Strategy Roadmap

	Predictable (Deterministic)	Uncertain (Stochastic)
Single-Period (One-Shot Bet)	**Trivial Exact Match** Order exact known demand $Q = D$. No leftovers, no stockouts.	**Module 3: Newsvendor Model** Fashion, tech launches, vaccines, fresh food. Balance underage (C_u) vs. overage (C_o): $\mathbf{P^* = \frac{C_u}{C_u + C_o}}$
Multi-Period (Ongoing)	**Module 2: EOQ Model** Stable, recurring industrial/retail SKUs. Balance setup (S) vs. holding (H): $\mathbf{Q^* = \sqrt{\frac{2DS}{H}}}$, $R = \bar{d}L$	**Modules 4 & 5: (Q, R) & Pooling** Ongoing replenishment under volatility. $\mathbf{Q^*}$ = EOQ, $\mathbf{R^* = \mu_{LTD} + z\sigma_{LTD}}$ Network Risk Pooling ($\sqrt{N}$ Law)

Inventory Costs: The Fundamental Batching Tension

Holding (Carrying) Cost (H)

Cost to keep **one unit** in inventory for **one year** (often expressed as % $\times$ unit cost P).
Pushes toward small, frequent orders ($Q \downarrow$).

Ordering / Setup Cost (S)

Fixed cost incurred **every time we place an order** or change over a line (independent of Q).
Pushes toward huge, infrequent batches ($Q \uparrow$).

The Batching Optimization Question

How large should our order quantity Q be to strike the exact minimum-cost balance between H and S?

Economic Order Quantity (EOQ): Scope & Assumptions

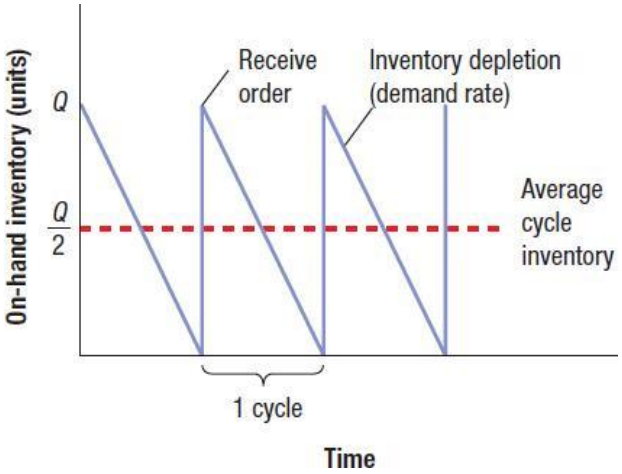
Economic Order Quantity (EOQ): The fixed lot size Q^* that minimizes total annual cycle-inventory holding and ordering costs.

Five Baseline Assumptions (And How We Will Relax Them)

1. **Constant, known demand rate (D):** Relaxed in Modules 3 & 4 when demand is random ($\sigma_d > 0$).
2. **Constant, known lead time (L):** Relaxed in Module 4 when supplier transit times vary ($\sigma_L > 0$).
3. **No batch constraints:** Unconstrained lot size Q (we will test pallet rounding sensitivity shortly!).
4. **Two relevant operational costs:** Annual unit holding cost (H) and fixed setup/order cost (S).
5. **Constant unit purchase price (P):** No quantity discounts; annual purchase cost $P \cdot D$ is fixed.

Visualizing Cycle Inventory: The Sawtooth Curve

Inventory Depletion and Instantaneous Replenishment of Lot Size Q



Building the Total Annual Cycle-Inventory Cost Function

1. Annual Holding Cost (Linear in Q)

$$\text{Annual Holding Cost} = (\text{Average Cycle Inventory}) \times (\text{Unit Holding Cost}) = \mathbf{\frac{Q}{2} \cdot H}$$

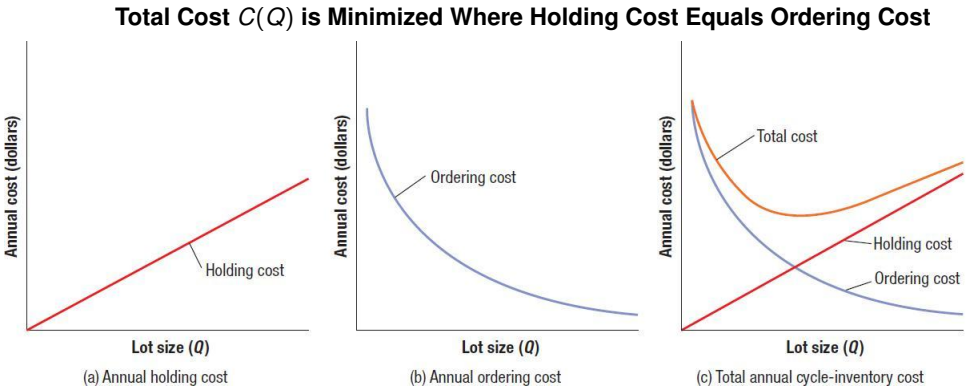
2. Annual Ordering / Setup Cost (Inverse in Q)

$$\text{Annual Ordering Cost} = \left(\text{Orders per Year} = \frac{D}{Q}\right) \times (\text{Cost per Order}) = \mathbf{\frac{D}{Q} \cdot S}$$

Total Annual Cycle-Inventory Cost Function $C(Q)$

$$C(Q) = \underbrace{\frac{Q}{2}H}_{\text{Holding Cost}} + \underbrace{\frac{D}{Q}S}_{\text{Ordering Cost}}$$

Visualizing the Cost Trade-off & Optimality Condition



The Economic Order Quantity (Q^*) Formula

Finding the Minimum of $C(Q)$

- First-order condition ($\frac{dC}{dQ} = 0$) or cost balance:

$$\frac{Q}{2}H = \frac{D}{Q}S$$

- Multiply by $\frac{2Q}{H}$:

$$(Q^*)^2 = \frac{2DS}{H} \implies \mathbf{Q^* = \sqrt{\frac{2DS}{H}}}$$

EOQ & Time Between Orders

$$Q^* = \sqrt{\frac{2DS}{H}}$$

$$\textbf{TBO} = \frac{Q^*}{D} \textbf{ years}$$

Orders per Year: $N = \frac{D}{Q^*}$

Cycle Time: TBO $\times$ 52 weeks (or 12 mos)

Diagnostic Rule for Executives

At Q^*, **Annual Holding Cost** = **Annual Ordering Cost**. If your warehouse holding bill is $10\times$ your annual ordering bill, your lot size Q is far too large!

Mini-Case 2: Auditing a Replenishment Policy

Pair Calculation & Poll (4 Minutes)

Formula Reference: $C(Q) = \frac{Q}{2}H + \frac{D}{Q}S$ | $Q^* = \sqrt{\frac{2DS}{H}}$ | $D = \bar{d} \times 52, \quad H = i \times P$

Operational Audit Context

- Specialty retail distribution SKU operating **52 weeks/year**.
- Current purchasing policy orders $Q = 468$ **units** (twice a year).
- A junior analyst proposes cutting the batch to $Q = 390$ **units**.

SKU Cost & Demand Parameters

- Weekly Demand ($\bar{d}$): **18 units/week** ($\implies D = 936$ units/yr)
- Unit Purchase Cost (P): **\$60 / unit**
- Fixed Ordering Cost (S): **\$45 / order**
- Holding Rate: **25% / yr** ($\implies H = \$15$/unit/yr)

Your Audit Questions:

1. Compare the total annual cost $C(Q)$ of $Q = 468$ vs. the analyst's $Q = 390$.
2. What is the **true EOQ (Q^*)** and its annual cost $C(Q^*)$?
3. **Poll:** If the supplier forces full pallets of $Q = 100$ instead of $Q^* = 75$, does annual cost jump by $< 5\%$ or $> 25\%$?

Executive Takeaways from EOQ: Robustness & The Square Root Law

1. Cost Curve Flatness

Because $C(Q)$ is flat near Q^*, rounding order quantities to convenient **pallet layers, cartons, or truckloads** (within $\pm 20\%$ of Q^*) increases total cost by $< 2\%$.

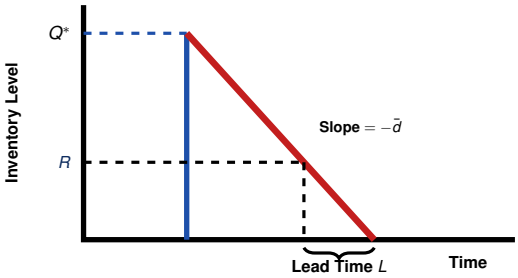
2. Square Root Law

Cycle stock $\frac{Q^*}{2} \propto \sqrt{D}$.
If product demand grows by $\mathbf{4}\times$, optimal cycle inventory grows by only $\sqrt{\mathbf{4}} = \mathbf{2}\times$ — doubling your inventory turns!

3. The Lean / JIT Lever

In Toyota Production System (Lean), S is not a fixed constant. Driving setup cost $\mathbf{S} \to \mathbf{0}$ (via EDI, quick die changes) allows optimal batch size $\mathbf{Q}^* \to \mathbf{1}$.

When to Order: Reorder Point (R) & Inventory Position



Deterministic Reorder Point (R)

$$\mathbf{R} = \mu_{\text{LTD}} = \bar{\mathbf{d}} \times \mathbf{L}$$

Crucial Distinction: Inventory Position

We trigger an order when **Inventory Position (IP)** hits R:

$$\text{IP} = \text{On-Hand} + \text{On-Order} - \text{Backorders}$$

(Prevents double-ordering when $L > TBO$!)

Quick Check: Deterministic Reorder Point (R)

Distribution Center Parameters

- Annual Demand (D): **8,000 units/year** across **250 working days/year**
- Supplier Lead Time (L): Typically **3 working days**, occasionally **4 working days**

Step 1: Compute Daily Demand Rate ($\bar{d}$)

$$\bar{d} = \frac{D}{\text{Working Days}} = \frac{8{,}000}{250} = \textbf{32 units/day}$$

Step 2: Compute Reorder Point ($R = \bar{d} \times L$)

- For $L = 3$ days: $R = 32 \times 3 = \textbf{96 units}$ (in Inventory Position)
- For $L = 4$ days: $R = 32 \times 4 = \textbf{128 units}$ (in Inventory Position)

Transitioning: From EOQ to the Newsvendor Model

Moving from multi-period deterministic batching to single-period betting under uncertainty

EOQ Model: Replenishment Logic

Balances **holding** (H) vs. **ordering** (S) costs.

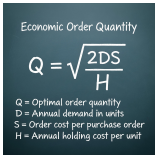
- Ongoing, non-perishable product
- Leftover inventory carries over to next week
- Question: *How frequently to batch orders?*

Newsvendor Model: Single-Period Bet

Balances **underage** (C_u) vs. **overage** (C_o) costs.

- Short selling season or perishable product
- Long lead time $\implies$ one commitment before seeing demand
- Question: *How much stock to bet on the season?*

The Trade-off:
Too Little (C_u) vs. Too Much (C_o)



The Consequence:
Lost Margin vs. Liquidation Loss

Industry Guide: Where Do You Use EOQ vs. Newsvendor?

Rule of Thumb: Can unsold inventory sit on the shelf and be sold next month at full price? **Yes** $\rightarrow$ **EOQ.** **No** $\rightarrow$ **Newsvendor.**

EOQ Use Cases (Long Shelf Life, Repeat Orders)

- **Consumer Packaged Goods (P&G, Unilever, Costco):** Paper towels, canned food, detergents, basic staples.
- **Industrial & MRO Distribution (Grainger, Fastenal, Caterpillar):** Fasteners, bearings, spare parts, raw steel/chemicals.
- **Healthcare & Hospital Supplies (McKesson, Cardinal Health):** Syringes, surgical gloves, saline, OTC medications.
- **Operational Signature:** Multi-year product lifecycle; replenishment lead time $L \ll$ product shelf life.

Newsvendor Use Cases (Perishable or One-Shot)

- **Apparel & Seasonal Retail (Zara, Nike, Lululemon):** Winter ski jackets, fashion capsules, Super Bowl championship merch.
- **Tech Hardware Launches (Apple, NVIDIA, Sony):** Initial production run for iPhone launch or holiday gaming consoles.
- **Fresh Food & Biotech (Whole Foods, Panera, Pfizer):** Fresh bakery/produce, seasonal flu vaccines, blood banks.
- **Operational Signature:** Supplier lead time $L >$ selling season (only **one commitment** before demand is revealed).

Module 3: Single-Period Inventory Strategy

Betting on Uncertain Demand

The Newsvendor Model & The Critical Fractile

Single-Period Betting: The Two Economic Risks

You must commit to an order quantity Q *before* observing random seasonal demand D

Under-Stocking Risk ($D > Q$)

Unit Underage Cost (C_u):

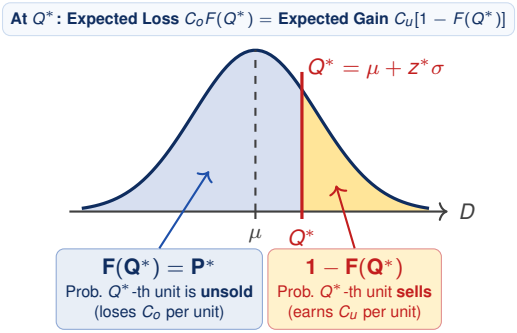
- You run out while customers are still buying.
- Every missed sale forfeits the full contribution margin $\mathbf{C_u = p - c}$ (plus customer goodwill).

Over-Stocking Risk ($D < Q$)

Unit Overage Cost (C_o):

- Season ends with unsold units on the shelf.
- Every leftover unit must be salvaged or liquidated at $s < c$, losing $\mathbf{C_o = c - s}$.

Intuition of the Newsvendor Model: Marginal Analysis



Why Not Just Order Average Demand (μ)?

Ordering $Q = \mu$ splits the curve 50–50 ($F(\mu) = 0.50$). That is only optimal if missing a sale (C_u) and having a leftover (C_o) cost the exact same!

Thinking at the Margin (The Q-th Unit)

- **Left of Q^* ($D \leq Q$):** The Q-th unit goes unsold with probability $F(Q)$, losing $\mathbf{C_o} \cdot \mathbf{F(Q)}$.
- **Right of Q^* ($D > Q$):** The Q-th unit sells with probability $1 - F(Q)$, earning $\mathbf{C_u} \cdot [\mathbf{1} - \mathbf{F(Q)}]$.
- **Optimal Cutoff (Q^*):** Stop adding units right where the two expected values balance!

Economic Optimization: The Critical Fractile & F^{-1}

1. Economic Costs & Service Level (P^*)

- **Underage Cost:** $\mathbf{C_u = p - c}$ (lost margin)
- **Overage Cost:** $\mathbf{C_o = c - s}$ (markdown loss)
- **Demand CDF** $F(Q) = \Pr(D \leq Q)$**:** The probability that demand does not exceed Q (your **in-stock service level**).
- **Critical Fractile Condition:**

$$\mathbf{F(Q^*) = P^* = \frac{C_u}{C_u + C_o}}$$

2. Solving for Q^* Requires Inverting F (Numerical Step!)

To get the order quantity Q^* from the target probability P^*, we must take the **inverse CDF**:

$$\mathbf{Q^* = F^{-1}(P^*)}$$

- **No Algebraic Formula:** For Normal demand $D \sim \mathcal{N}(\mu, \sigma^2)$, F^{-1} cannot be solved with algebra—it is a **numerical lookup step**.
- **Standard Normal CDF** $\Phi(z) = \Pr(Z \leq z)$**:**

$$\mathbf{z^* = \blacksquare^{-1}(P^*)} \quad \Longrightarrow \quad \mathbf{Q^* = \mu + z^*\sigma}$$

Find z^* via a **Standard Normal Table** or Excel `=NORM.S.INV(`P^*`)`.

How to Use the Critical Fractile: 3-Step Numerical Recipe

Walkthrough Example: Suppose $\mu = 200$, $\sigma = 30$, $C_u = \$40$, and $C_o = \$10$:

Step 1: Compute Target Service Level (P^*)

$$P^* = \frac{C_u}{C_u + C_o} = \frac{40}{40 + 10} = \mathbf{0.80} \quad \text{(target 80\% in-stock probability)}$$

Step 2: Numerical Inversion $z^* = \Phi^{-1}(P^*)$

Find the z-score where cumulative standard normal area $\Phi(z^*) = 0.80$:

- **By Table (Right):** Locate **0.80** in the $\Phi(z)$ column $\longrightarrow$ read across to get $\mathbf{z^* = 0.842}$.
- **By Excel:** Type `=NORM.S.INV(0.80)` $\longrightarrow$ returns **0.842**.

Step 3: Convert z^* into Units ($Q^* = F^{-1}(P^*)$)

$$Q^* = \mu + z^*\sigma = 200 + (0.842)(30) = 200 + 25.3 \approx \mathbf{225 \text{ units}}$$

Standard Normal Lookup

Invert $\Phi(z) = P^*$ to read $z^* = \Phi^{-1}(P^*)$

Prob. $\Phi(z) = P^*$	$z^* = \Phi^{-1}(P^*)$
0.50	0.000
0.60	0.253
0.70	0.524
0.75	0.674
0.80	$\longrightarrow$ **0.842**
0.85	1.036
0.90	1.282

Direct 1-Step Excel Shortcut:

`=NORM.INV(0.80, 200, 30)` = **225.3**

Mini-Case 3: Daily Artisan Bakery Production

Pair Calculation & Poll (4 Minutes)

Formula Reference: $C_u = p - c$ | $C_o = c - s$ | $P^* = \dfrac{C_u}{C_u + C_o}$ | $z^* = \Phi^{-1}(P^*)$ [Table] | $Q^* = \mu + z^*\sigma$

Standard Normal Table

$\Phi(z) = P^*$	$z^* = \Phi^{-1}(P^*)$
0.50	0.000
0.55	0.126
0.60	0.253
0.65	0.385
0.70	0.524
0.75	0.674
0.80	0.842
0.85	1.036

Excel: `=NORM.S.INV(P*)`

Artisan Bakery Parameters

- Daily Demand: Normal with $\mu = \mathbf{90}$ **loaves**, $\sigma = \mathbf{10}$ **loaves**
- Selling Price: $\mathbf{p} = \$\mathbf{4.50}$ — Baking Cost: $\mathbf{c} = \$\mathbf{1.50}$
- Day-Old Salvage Value: $\mathbf{s} = \$\mathbf{0.50}$

Your Executive Challenge:

1. Compute the underage cost C_u, overage cost C_o, and optimal service level P^*.
2. Use the table to invert $z^* = \Phi^{-1}(P^*)$ and find the optimal daily baking quantity Q^*.
3. **Bonus Poll:** Under what cost condition should a business intentionally stock *less* than average demand μ ($P^* < 50\%$)?

Newsvendor Comparative Statics: Propensity to Hold Inventory

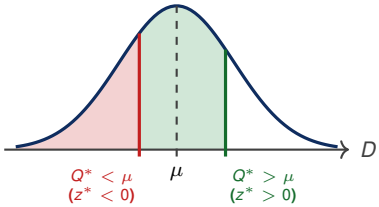
How changing economic parameters shifts your optimal service level $P^* = \frac{C_u}{C_u + C_o}$ and order size Q^*:

Parameter Shift	Effect on C_u and C_o	P^*	Q^*	Managerial Intuition
Selling Price ($p \uparrow$) (or Gross Margin ↑)	$C_u = p - c$ **rises**	↑	↑	Missing a sale is more painful → **higher propensity to stock**.
Unit Cost ($c \uparrow$) (holding p, s fixed)	$C_u \downarrow$ **and** $C_o = c - s \uparrow$	↓	↓	Smaller margin & bigger markdown loss → **order more conservatively**.
Salvage Value ($s \uparrow$) (or Supplier Buyback)	$C_o = c - s$ **falls**	↑	↑	Leftovers are cheaper to liquidate → **willing to bet bigger**.
Carrying / Disposal ($h \uparrow$) (cost per leftover unit)	$C_o = c - s + h$ **rises**	↓	↓	Carrying or disposing of leftovers is costlier → **shrink order size**.

Key Case-Study Rule of Thumb

Any lever that **increases the reward of selling ($p \uparrow$)** or **cushions the penalty of leftovers ($s \uparrow$)** pushes P^* and Q^* **up**.
Any lever that **inflates unit cost ($c \uparrow$) or carrying/disposal cost ($h \uparrow$)** pushes P^* and Q^* **down**.

When Do You Stock Above vs. Below Average Demand (μ)?



1. Stock ABOVE Average ($Q^* > \mu$)

- **Condition: $C_u > C_o \iff P^* > 50\% \iff z^* > 0$.**
- Lost margin exceeds leftover loss $\rightarrow$ hold **positive safety stock** above μ.

2. Stock BELOW Average ($Q^* < \mu$)

- **Condition: $C_u < C_o \iff P^* < 50\% \iff z^* < 0$.**
- Leftovers destroy more value than a stockout $\rightarrow$ **intentionally ration supply** below μ!

3. Surprising Effect of Uncertainty ($\sigma \uparrow$)

Since $Q^* = \mu + z^*\sigma$, higher volatility ($\sigma \uparrow$) **increases** Q^* when $P^* > 50\%$ ($z^* > 0$), but **decreases** Q^* **even further** when $P^* < 50\%$ ($z^* < 0$)!

Industry Guide: Who Stocks Above μ vs. Below μ?

Stock Well ABOVE Average ($P^* > 50\%,\ Q^* > \mu$)

Economics: High gross margins (C_u huge), low unit cost or high salvage (C_o small).

- **Biotech & Pharmaceuticals:** Unit cost $c = \$20$, price $p = \$500$ ($C_u = \$480 \gg C_o = \$20 \implies \mathbf{P^* = 96\%}$).
- **Luxury Apparel & Cosmetics (LVMH):** 80%+ gross margins; running out of a hit handbag forfeits massive margin.
- **Print Newspapers / Coffee Shops:** A cup of coffee or paper costs pennies to make (C_o tiny) and sells for 4× cost ($P^* \approx 75$–85%).

Stock Intentionally BELOW Average ($P^* < 50\%,\ Q^* < \mu$)

Economics: Thin margins (C_u small), high wholesale cost & total spoilage/obsolescence (C_o huge).

- **High-End Fresh Sushi / Raw Seafood:** Wholesale $c = \$15$/lb, price $p = \$19$ ($C_u = \$4$), zero overnight salvage ($C_o = \$15 \implies \mathbf{P^* = 21\%}$). Selling out early is optimal!
- **Custom / Bespoke Hardware & Servers:** Expensive build cost ($c = \$2{,}000$), modest markup ($C_u = \$400$), steep write-off if unsold ($C_o = \$1{,}200 \implies \mathbf{P^* = 25\%}$).
- **Holiday Cut Flowers / Fresh Berries:** High purchase cost and 100% spoilage the day after Valentine's Day.

Summary: Optimizing the Newsvendor Policy

Three-step recipe for any single-period stocking decision under uncertainty:

01. Quantify Economic Stakes

Underage Cost: $C_u = p - c$ (lost margin per stockout).
Overage Cost: $C_o = c - s$ (loss per leftover unit).

02. Compute Critical Fractile (P^*)

Optimal cycle service level (probability of meeting all demand):

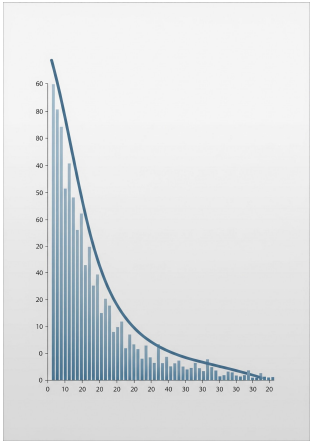
$$P^* = \frac{C_u}{C_u + C_o}$$

03. Translate P^* to Order Quantity (Q^*)

Find z^* where $\Phi(z^*) = P^*$ and set $\mathbf{Q}^* = \mu + \mathbf{z}^*\sigma$.

Handling Non-Normal & Empirical Demand Distributions

The Critical Fractile $P^* = \frac{C_u}{C_u + C_o}$ holds for *any* demand distribution (discrete or empirical):



Rule for Discrete / Empirical Demand

When demand comes in whole units, you cannot hit P^* exactly. Pick the **smallest integer** Q^* where the cumulative probability $F(Q)$ first reaches or exceeds P^*:

$$\mathbf{F(Q^*) = \Pr(D \leq Q^*) \geq P^*}$$

Trivial Example: Spare Turbines (Target $P^* = 0.75$)

Demand (Q)	0	1	**2**	3
Prob. $\Pr(D = Q)$	0.10	0.40	0.30	0.20
Cumul. $F(Q) = \Pr(D \leq Q)$	0.10	0.50	**0.80 $\geq$ 0.75**	1.00

Decision: $F(1) = 0.50 < 0.75$ is too low; $\mathbf{Q^* = 2}$ is the first quantity where $F(Q) \geq 0.75 \implies$ **Stock $\mathbf{Q^* = 2}$ units**.

The "Data-Driven Newsvendor" Debate: Is the Model Obsolete?

The Analytics Claim: "Infer Q^* Directly from Data"

A major movement in prescriptive analytics (e.g., at MIT) argues against fitting Normal distributions $\mathcal{N}(\mu, \sigma^2)$:

- **Skip μ, σ, and Φ^{-1}:** Sort your N historical demand observations $D_{(1)} \leq D_{(2)} \leq \cdots \leq D_{(N)}$ (or run quantile regression on features X).
- **Direct Empirical Illustration ($N = 10$, $P^* = 0.75$):**

  $$\{60,\ 68,\ 72,\ 79,\ 84,\ 88,\ 92,\ \mathbf{96},\ 105,\ 118\}$$

  Pick the $\lceil 0.75 \times 10 \rceil = 8$th sorted value
  $\Longrightarrow$ $\mathbf{Q}^* = \mathbf{96}$ directly from data—no bell curve needed!

The Verdict: What Is Obsolete (and What Isn't!)

- **What *Is* Obsolete:** Blindly forcing skewed or multi-modal data into a textbook $\mathcal{N}(\mu, \sigma^2)$ bell curve.
- **Why Newsvendor Theory Is *Not* Obsolete:**
  1. **Economics Still Picks the Percentile:** Without the Newsvendor Critical Fractile $P^* = \frac{C_u}{C_u + C_o}$, the data cannot tell you whether to pick the 2nd or 8th observation!
  2. **Small Data & Censored Stockouts:** For new launches ($N < 15$) or when past stockouts hid true demand, pure "model-free" data overfits badly.

Strategic Levers: How Executives Beat the Newsvendor Curse

Don't Just Accept σ and C_o — Redesign the Supply Chain!

Top operations executives actively alter the parameters of the Newsvendor problem:

1. Quick Response ($\sigma \downarrow$)

Example: Zara
Cut design-to-rack lead times from 6 months to 15 days. Ordering closer to the selling season slashes forecast error σ.

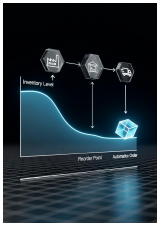
2. Postponement

Example: Benetton / HP
Hold generic "greige" sweaters or unlocalized printers; dye/customize only after early demand signals arrive.

3. Buyback Contracts

Example: Publishing / Tech
Suppliers agree to buy back unsold units at $b > s$, reducing the retailer's overage cost C_o and lifting Q^*.

Moving to Multi-Period Uncertainty: The (Q, R) Policy



The Continuous-Review (Q, R) Policy

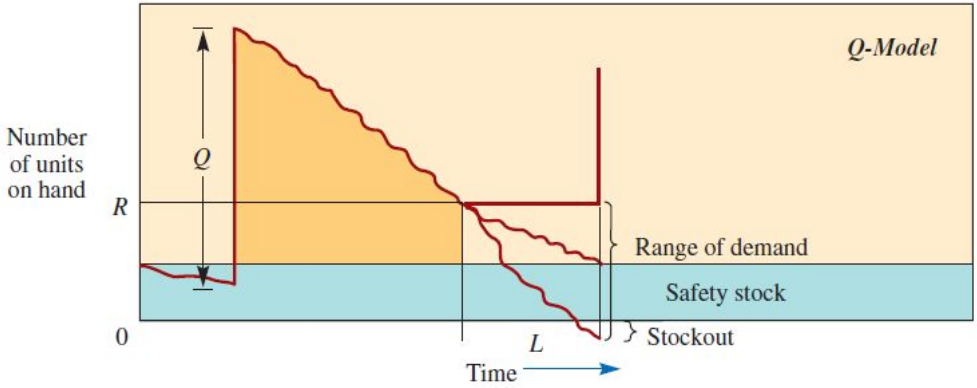
- Track **Inventory Position (IP)** continuously in real time.
- Whenever IP drops to the **Reorder Point (R)**, place a fixed order of size **Q**.
- Designed for ongoing SKUs facing random demand (σ_d) and/or random supplier lead times (σ_L).

Average On-Hand Inventory in a (Q, R) System

$$\text{Average On-Hand Inventory} \approx \underbrace{\frac{Q}{2}}_{\text{Cycle Stock}} + \underbrace{SS}_{\text{Safety Stock}}$$

Visualizing the Stochastic (Q, R) Trajectory

Crucial Insight: Stockouts Only Occur During the Replenishment Lead Time (L)!



Decoupling the Two Decisions: Order Size (Q^*) and Reorder Point (R^*)

Decision 1: How Much to Order (Q^*) $\rightarrow$ EOQ

$$\mathbf{Q}^* = \sqrt{\frac{\mathbf{2DS}}{\mathbf{H}}}$$

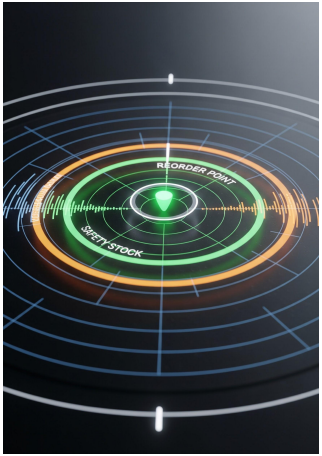
Driven by **average annual demand (*D*)**, setup cost (S), and holding cost (H). Because the EOQ cost curve is flat, using EOQ for Q^* is a near-exact approximation under uncertainty!

Decision 2: When to Order (R^*) $\rightarrow$ Safety Stock

$$\mathbf{R}^* = \mu_{\text{LTD}} + \underbrace{z \cdot \sigma_{\text{LTD}}}_{\text{Safety Stock (SS)}}$$

All demand and lead-time uncertainty (σ_{LTD}) is absorbed by the **Reorder Point (R^*)**, not by inflating Q^*!

Level 1 Uncertainty: Random Demand, Constant Lead Time (L)



Aggregating Demand Over L Independent Periods

- **Expected Lead-Time Demand (μ_{LTD}):**

$$\mu_{\text{LTD}} = \bar{\mathbf{d}} \cdot \mathbf{L}$$

- **Std. Dev. of Lead-Time Demand (σ_{LTD}):** Variances add across L independent days/weeks ($\sigma^2_{\text{LTD}} = L\sigma^2_d$):

$$\sigma_{\text{LTD}} = \sigma_{\mathbf{d}}\sqrt{\mathbf{L}}$$

Reorder Point & Safety Stock Formula

$$\text{SS} = \mathbf{z}\sigma_{\mathbf{d}}\sqrt{\mathbf{L}}, \qquad \mathbf{R}^* = \bar{\mathbf{d}}\mathbf{L} + \mathbf{z}\sigma_{\mathbf{d}}\sqrt{\mathbf{L}}$$

Quick Warm-Up: Regional Fulfillment Center

Formula Reference: $\mu_{\text{LTD}} = \bar{d} \cdot L$ | $\sigma_{\text{LTD}} = \sigma_d \sqrt{L}$ | $\text{SS} = z \cdot \sigma_{\text{LTD}}$ | $R^* = \mu_{\text{LTD}} + \text{SS}$

E-Commerce Fulfillment Center Parameters

A regional fulfillment center replenishes a popular consumer electronics accessory with a **constant supplier lead time of** $L = 10$ **days**. Daily demand is normally distributed with:

- Mean daily demand: $\bar{\mathbf{d}} = \mathbf{1{,}250}$ **units/day**
- Standard deviation of daily demand: $\sigma_{\mathbf{d}} = \mathbf{375}$ **units/day**
- Target Cycle Service Level: **90%** ($\implies$ $\mathbf{z = 1.28}$)

2-Minute Warm-Up Calculation:

1. What is the Expected Lead-Time Demand (μ_{LTD})?
2. What is the standard deviation of Lead-Time Demand (σ_{LTD})?
3. What Safety Stock (SS) and Reorder Point (R^*) achieve the 90% service target?

Mini-Case 4: Julia's Mercedes-AMG Dealership

Formula Reference: $Q^* = \sqrt{\frac{2DS}{H}} \mid \mu_{\text{LTD}} = \bar{d} \cdot L \mid \sigma_{\text{LTD}} = \sigma_d\sqrt{L} \mid \text{SS} = z \cdot \sigma_{\text{LTD}} \mid R^* = \mu_{\text{LTD}} + \text{SS}$

Dealership Parameters (Guaranteed Factory Lead Time)

- **Weekly Demand:** $\bar{\mathbf{d}} = 5$ **cars/wk**, $\sigma_{\mathbf{d}} = 3$ **cars/wk** ($D = 250$ cars/yr)
- **Factory Lead Time:** Guaranteed $\mathbf{L} = 4$ **weeks** ($\sqrt{4} = 2$)
- **Costs & Service:** $\mathbf{S} = \$1{,}800$, $\mathbf{H} = \$9{,}000$/yr, **95% CSL** ($\mathbf{z} = 1.65$)

Julia's Puzzle: "Why Do We Keep Stocking Out?!"

Julia orders $\mathbf{Q} = 25$ **cars** whenever inventory hits $\mathbf{R} = 20$ **cars** (the 4-week average demand). She pays huge floorplan interest yet stocks out 50% of the time!

1. Find her optimal order size $\mathbf{Q}^*$ and optimal reorder point $\mathbf{R}^*$.
2. **Bonus:** If the factory guarantees **1-week delivery** ($L = 1$ **wk**), how many Safety Stock cars and how many dollars/year does she save?

[Advanced / Optional] When Both Demand & Lead Time Vary

Optional Reference Material — Not Required for In-Class Exercises

Two Independent Buckets of Supply Chain Risk

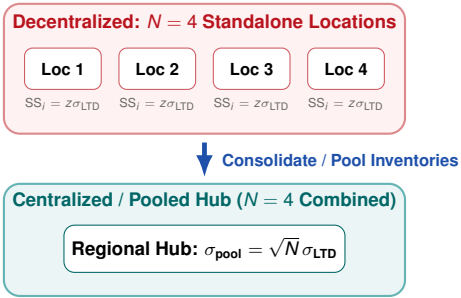
If supplier lead time is also random with mean $\bar{L}$ and std. dev. σ_L (and orders do not cross):

$$\sigma_{\text{LTD}} = \sqrt{\underbrace{\bar{L}\sigma_d^2}_{\substack{\text{Bucket 1:}\\ \text{Demand Volatility}}} + \underbrace{\bar{d}^2\sigma_L^2}_{\substack{\text{Bucket 2:}\\ \text{Supplier Delay Risk}}}}$$

Strategic Takeaway for Operations Executives

Notice that supplier lead-time volatility (σ_L^2) is multiplied by $\bar{\mathbf{d}}^{\mathbf{2}}$: for high-volume SKUs, **unreliable supplier delivery times** inflate safety stock far more than customer demand fluctuations!

Supply Chain Network Strategy: Risk Pooling & The Square Root Law



Why Risk Pooling Works

When N independent locations with identical demand volatility σ_{LTD} pool their inventory, high demand in one market offsets low demand in another:

$$\sigma_{\text{pooled}} = \sqrt{\sigma_1^2 + \cdots + \sigma_N^2} = \sqrt{\mathbf{N}} \cdot \sigma_{\text{LTD}}$$

The Square Root Law of Safety Stock

$$\text{SS}_{\text{pooled}} = \frac{\text{SS}_{\text{decentralized}}}{\sqrt{\mathbf{N}}}$$

Risk Pooling in Action: Consolidating 4 Dealerships / Warehouses

1. Four Separate Dealerships ($N = 4$)

Suppose Julia and 3 neighboring AMG dealerships each hold their own safety stock ($SS_i = 10$ cars for 95% CSL):

- **Per-Dealer Safety Stock:** 10 cars ($\sigma_{LTD} = 6$)
- **Total Network SS:** $4 \times 10 = \mathbf{40}$ **cars**
- **Annual Holding Cost of SS:** $40 \times \$9{,}000 = \$\mathbf{360{,}000}/\text{yr}$

2. One Pooled Regional Hub ($N = 4$)

The 4 dealers share a single regional inventory pool (or virtual inventory transshipment network):

- **Pooled Std. Dev.:** $\sqrt{4} \times 6 = 12$ cars
- **Total Network SS:** $\frac{40}{\sqrt{4}} = 1.65(12) \approx$ **20 cars**
- **Result: 50% less safety stock** (saves $\$\mathbf{180{,}000}/\text{yr}$) at the **exact same 95% service level**!

Strategic Trade-Off: Inventory Capital vs. Last-Mile Response Time

Centralization slashes safety stock by $1 - \frac{1}{\sqrt{N}}$, but increases physical distance to the customer. **Virtual pooling** (e.g., dealer inventory sharing, ship-from-store) captures pooling benefits while keeping stock close to customers!

Practical Implementation: Review Systems & ABC Segmentation

Continuous (Q, R) vs. Periodic Review

- **Continuous Review (Q, R):** Order fixed quantity Q whenever IP $\leq R$. Best for high-value items with real-time ERP tracking.
- **Periodic Review (T, S):** Check stock every fixed T days (e.g., weekly truck delivery) and order up to target level S. Requires slightly more safety stock (buffers $T + L$), but coordinates multi-item shipments.

ABC Pareto Inventory Segmentation

Not all SKUs deserve the same analytical rigor:

- **Class A ($\sim$ 20% of SKUs, 80% of $ value):** Tight continuous (Q, R) control, frequent lead-time audits, executive oversight.
- **Class B ($\sim$ 30% of SKUs, 15% of $ value):** Automated EOQ / periodic review with standard service targets.
- **Class C ($\sim$ 50% of SKUs, 5% of $ value):** Simple two-bin systems or generous bulk buffers so cheap fasteners never halt assembly.

1960s Theory Meets 2026 AI: Did AI Replace These Models?

The Short Answer: AI Does Not Repeal the Laws of Economics!

Whether a human planner or a deep neural network places the order, the fundamental trade-offs of inventory theory remain unchanged: **Setup (S) vs. Holding (H)**, **Underage (C_u) vs. Overage (C_o)**, and **Little's Law ($\bar{d} \times L$)**.

What 1960s Classical Models Assumed

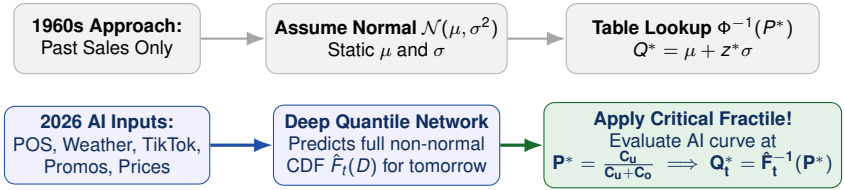
- **Static Parameters:** μ, σ_d, and L are fixed constants updated once a year in an ERP spreadsheet.
- **Bell-Curve World:** Demand follows a neat, stationary Normal distribution $\mathcal{N}(\mu, \sigma^2)$.
- **Single-SKU Silos:** Each item and warehouse is optimized independently with closed-form algebra.

How 2026 AI Supercharges the Models

- **Dynamic Contextual Inputs:** ML predicts $\hat{\mu}_t, \hat{\sigma}_t, \hat{L}_t$ **every day** using hundreds of live signals (weather, promos, social trends, port delays).
- **Direct Quantile Prediction:** Deep learning predicts the entire curve $\hat{F}_t^{-1}(P^*)$ without assuming Normality.
- **Network-Wide RL:** Reinforcement learning coordinates millions of SKUs across multi-echelon networks.

AI Paradigm 1: Probabilistic Forecasting Meets the Critical Fractile

How modern retail AI (e.g., Amazon MQ-Transformer / DeepAR) directly uses the Newsvendor formula:



Why the 1960s Critical Fractile ($P^* = \frac{C_u}{C_u+C_o}$) Is More Important Than Ever

- A neural network can tell you the 50th, 75th, and 95th percentile of tomorrow's demand, **but it does not know which percentile to order** unless you give it the economic Critical Fractile $P^* = \frac{C_u}{C_u+C_o}$!
- Better AI features shrink the forecast error ($\sigma_{\text{error}} \downarrow$), allowing firms to hit the **same service level P^* with far less safety stock**.

AI Paradigm 2: Cost-Aware Deep Learning & Reinforcement Learning

1. The "Forecast Accuracy" Trap

Standard ML models minimize symmetric Mean Squared Error (MSE), treating $+10$ and -10 errors equally.

- **The Problem:** If missing a sale costs $C_u = \$40$ and a leftover costs $C_o = \$5$, an "unbiased" MSE forecast loses massive profit!
- **End-to-End (Decision-Focused) ML:** Train the neural network directly on the **Newsvendor Loss Function** (asymmetric pinball loss) so the AI outputs the profit-maximizing order Q_t^* directly from raw data.

2. Deep Reinforcement Learning (RL)

When does (Q, R) break down in practice?

- Multi-echelon networks (factories $\rightarrow$ DCs $\rightarrow$ 2,000 stores) with shared truck capacities, perishable shelf lives, and lost-sales censoring.
- **How RL Works:** A digital twin simulates millions of supply chain seasons; the RL agent learns a dynamic state-dependent policy $\pi(\text{Inventory}, \text{Pipeline}, \text{Context}) \rightarrow Q_t$ that outperforms static (Q, R) heuristics by 5–15%.

AI Paradigm 3: Lead-Time Sensing, Computer Vision & Agentic ERP

Beyond demand forecasting, AI attacks the operational friction inside (Q, R) execution:

1. Dynamic Lead-Time AI

- ERPs traditionally hardcode a static lead time (e.g., $L = 14$ days).
- ML ingests live port congestion, GPS/AIS ship tracking, weather, and supplier backlog data to update $\hat{L}_t$ and $\hat{\sigma}_{L,t}$ **in real time**, raising R_t^* *before* delays cause stockouts.

2. Fixing "Phantom Inventory"

- **The (Q, R) Blindspot:** If the ERP thinks IP $= 15$ (above $R^* = 10$), it never orders—even if theft or misplacement left the shelf at 0!
- **AI Computer Vision & RFID:** Shelf cameras, warehouse drones, and ML anomaly detection audit real-time physical stock so IP is accurate.

3. Agentic Supply Chain AI

- **2025–2026 Frontier:** LLM-powered ERP agents triage thousands of daily stockout alerts.
- Agents autonomously email suppliers to confirm lead times, propose cross-store transshipments (virtual pooling), and draft purchase orders for human sign-off.

Industry Frontier: What Leading Companies Are Actually Doing

Amazon: Deep Quantiles & Predictive Placement

- Replaced classic moving averages with **MQ-Transformer** neural networks predicting full demand distributions across 400M+ SKUs.
- Combines **Newsvendor logic** with **Multi-Echelon RL** to position inventory into regional fulfillment centers *before* customers click buy (enabling same-day delivery without exploding safety stock).

Walmart: AI Virtual Pooling Across 4,700 Stores

- Uses ML to dynamically route online orders to stores with excess inventory ($D < Q$, high overage risk C_o) while protecting inventory at stores likely to sell out locally (C_u).

Zara & Shein: AI-Driven "Quick Response"

- Instead of betting the whole season on one massive Newsvendor order Q^*, they launch tiny test batches and use real-time AI trend-sensing to trigger rapid in-season production on winners.
- **Theoretical Lesson:** Using speed + AI to slash σ_{LTD} beats holding huge safety buffers!

Sysco & Grocery Retailers: Perishable Fresh AI

- Combine daily Newsvendor ordering with **dynamic AI markdown pricing**: as expiration approaches, AI lowers price p to raise salvage value s and clear inventory.

Executive Playbook: Best Practices for Implementing AI in Inventory

How to succeed (and avoid multi-million-dollar AI failures) as an operations leader:

1. Fix Data & Inventory Accuracy First

"Garbage In, Garbage Out": Do not buy a $5M AI engine if your ERP lacks clean stockout flags (confusing *zero stock* with *zero demand*!) or has 25% phantom inventory errors.

2. Optimize Economics (C_u, C_o, H, S), Not MAPE

Force your data science team to evaluate models on **inventory working capital and lost-margin dollars** (the Critical Fractile), never on symmetric forecast error metrics alone.

3. Segment by ABC/XYZ (Don't Over-Engineer)

- **Stable / Low-Value SKUs:** Classical EOQ and (Q, R) achieve 99% of optimal performance at zero compute cost.
- **Volatile / High-Margin / Perishable SKUs:** Deploy probabilistic AI and dynamic quantile ordering where σ and C_u/C_o stakes are highest.

4. Enforce Hard Operational Guardrails

Wrap black-box AI models in classical constraints: pallet rounding (Slide 16), supplier MOQs, warehouse capacity, and human approval for outlier orders.

Master MBA Inventory Model Cheat Sheet

Model / Bucket	When to Use	Core Economic Trade-Off	Key Actionable Formula(s)
Pipeline Stock (Little's Law)	Any multi-stage supply chain with transit time L	Transit speed (L) vs. working capital tied up on water/road	Pipeline Stock $= R \cdot L$
EOQ Model (Cycle Stock)	Ongoing, predictable demand (D); batch replenishment	Fixed setup/order cost (S) vs. annual holding cost (H)	$Q^* = \sqrt{\frac{2DS}{H}}$, Cycle Stock $= \frac{Q^*}{2}$
Newsvendor (Single-Period)	Perishable goods, fashion capsules, one-shot launches	Underage lost margin ($C_u = p - c$) vs. overage loss ($C_o = c - s$)	$P^* = \frac{C_u}{C_u + C_o}$, $Q^* = F^{-1}(P^*) = \mu + z^*\sigma$
(Q, R) **Policy & Risk Pooling**	Ongoing SKUs with uncertain demand (σ_d) over lead time L	Cycle service level (z) vs. safety stock capital; centralization	$R^* = \bar{d}L + z\sigma_d\sqrt{L}$ $\text{SS}_{\text{pooled}} = \frac{\text{SS}_{\text{decentralized}}}{\sqrt{N}}$

Executive Takeaways for Managing Inventory

Strategic Synthesis for Operations Leaders

1. **Pipeline Stock Depends on Speed ($R \times L$):** Cutting transit lead times via nearshoring frees massive working capital even when suppliers require larger batches.
2. **EOQ is Forgiving Near Q^*:** Use $Q^* = \sqrt{2DS/H}$ to find the optimal zone, round freely to full pallets, and attack setup costs ($S \downarrow$) to enable Lean batches.
3. **Let the Critical Fractile Drive Single-Period Bets:** Use $P^* = \frac{C_u}{C_u + C_o}$ and numerical inversion $Q^* = F^{-1}(P^*)$ to decide whether to buffer above μ or ration below μ.
4. **Decouple Q^* and R^* Under Multi-Period Uncertainty:** Use Q^* (EOQ) for batch sizing and $R^* = \bar{d}L + z\sigma_d\sqrt{L}$ (via **Inventory Position**) to buffer lead-time volatility.
5. **Pool Risk ($\sqrt{N}$) & Deploy Cost-Aware AI:** Centralization slashes safety stock by $1/\sqrt{N}$, while modern AI supercharges classical models by predicting dynamic quantiles $\hat{F}_t^{-1}(P^*)$ and live lead times $\hat{L}_t$.